📘 Individualized Student Task: Unique Matrix Value Investigation

🎯 Objective

To analyze how **changing matrix values** affects matrix properties, using a **student-specific matrix**.

🧩 Task Rules (Important)

👉 **Each student must use a different matrix** generated from their own ID.

📝 Task Description

1. Create Your Personal Matrix

Let:

- d1 = last digit of your student ID
- d2 = second last digit of your student ID

Create the matrix:

$$A = \begin{bmatrix} d1 + 2 & d2 + 1 \\ 2d1 & d2 + 2 \end{bmatrix}$$

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2. Analyze Matrix A

Using NumPy, compute:

- Shape of A
- Determinant
- Rank
- Eigenvalues
- Inverse (only if determinant ≠ 0)

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3. Value Change Experiment

Create a new matrix **B** by changing **exactly one value** in A:

- Add **+1** to one element of your choice
- You must clearly state which element you changed

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4. Re-analyze Matrix B

Compute the same properties as in Step 2.

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5. Comparison & Explanation

Answer the following in your own words:

1. How did the determinant change and why?
2. Did the rank change? Explain.
3. How did the eigenvalues respond to the value change?
4. Is B easier or harder to invert than A? Why?